

For each problem, find the average value of the function over the given interval.

5) $f(x) = -x^3 + 7x^2 - 11x + 3$; $[1, 5]$

6) $f(x) = -x^5 + 3x^3$; $[0, 1]$

7) $f(x) = 4x^{\frac{1}{2}}$; $[0, 3]$

8) $f(x) = x^5 - 2x^3 + x$; $[-1, 0]$

9) $f(x) = \frac{1}{x}$; $[2, 3]$

10) $f(x) = x^5 - 4x^3 + 2x - 1$; $[-2, 2]$

11) $f(x) = -x^5 + 4x^3 - 5x - 3$; $[-2, 0]$

12) $f(x) = x^5 - 2x^3 - 2$; $[-1, 1]$

For each problem, find the average value of the function over the given interval. Then, find the values of c that satisfy the Mean Value Theorem for Integrals.

13) $f(x) = -x + 2$; $[-2, 2]$

14) $f(x) = -x^2 - 8x - 17$; $[-6, -3]$

15) $f(x) = -3(2x - 6)^{\frac{1}{2}}$; $[3, 5]$

16) $f(x) = \frac{4}{(2x + 6)^2}$; $[-6, -5]$